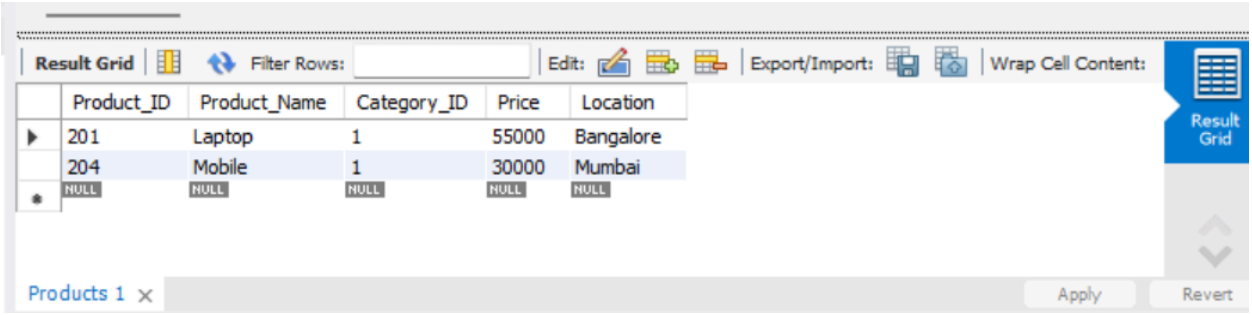
Subquery 1 — Products costing above the average

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Price > (SELECT AVG(Price) FROM Products);
```

OUTPUT:

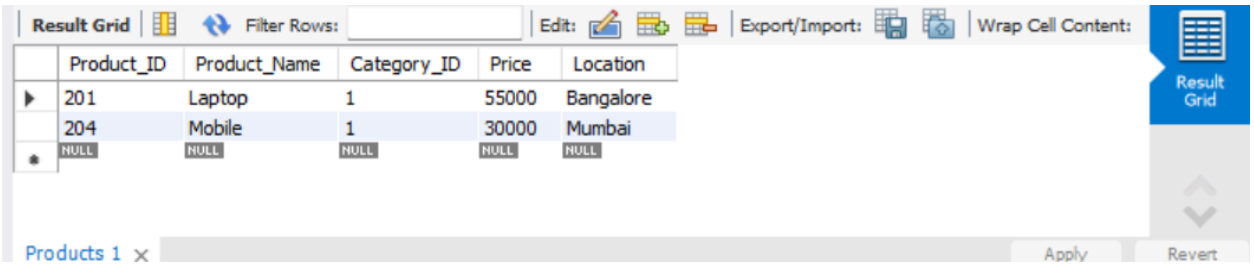


Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Outer:

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Price >
(SELECT AVG(Price) FROM Products);
```

OUTPUT:



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Result Grid

Products 1 ×

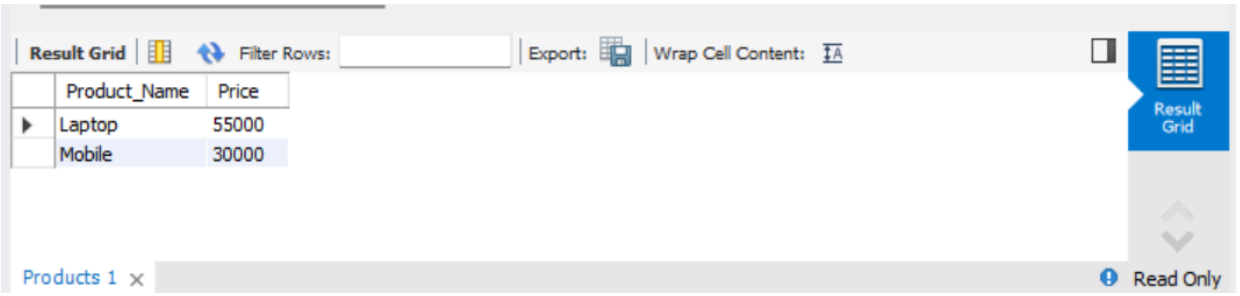
Apply Revert

Inner:

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_Name, Price
FROM Products
WHERE Price >
(SELECT AVG(Price) FROM Products);
```

OUTPUT:

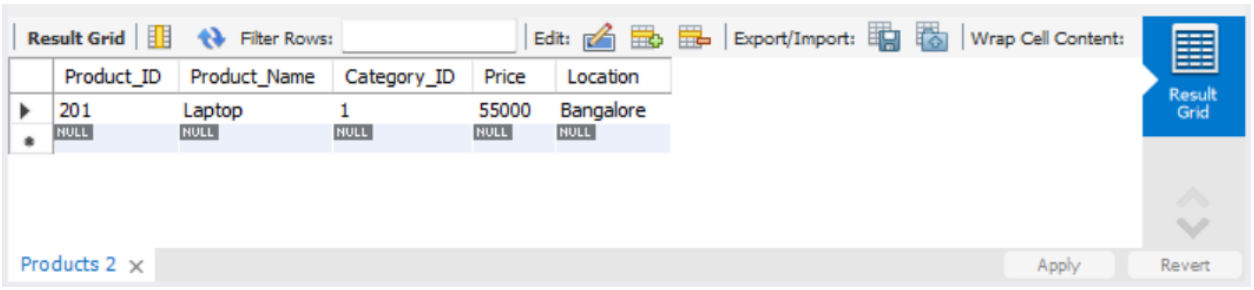


Subquery 2 — Product(s) with the highest price

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Price = (SELECT MAX(Price) FROM Products);
```

OUTPUT:



INNER:

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT MAX(Price)
FROM Products;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MAX(Price)
55000

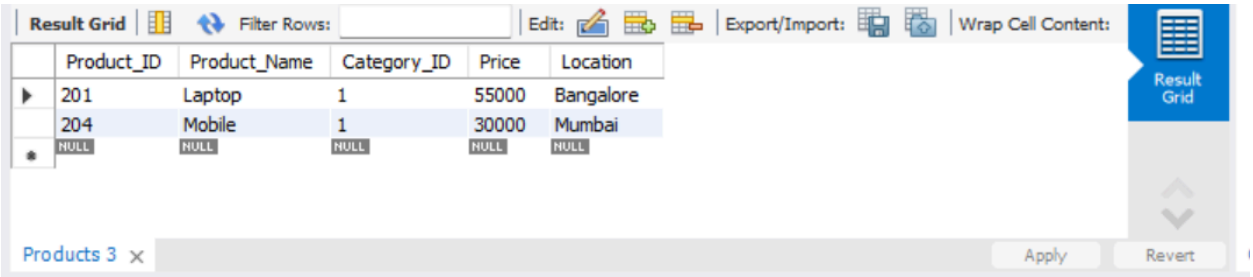
Result 1 | Read Only

Subquery 3 — Products costing above Laptop

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Price >
(SELECT Price FROM Products WHERE Product_Name = 'Table');
```

OUTPUT:



Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Subquery 4 — Products in the same category as Laptop

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Category_ID =
(SELECT Category_ID FROM Products WHERE Product_Name = 'Laptop');
```

OUTPUT:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

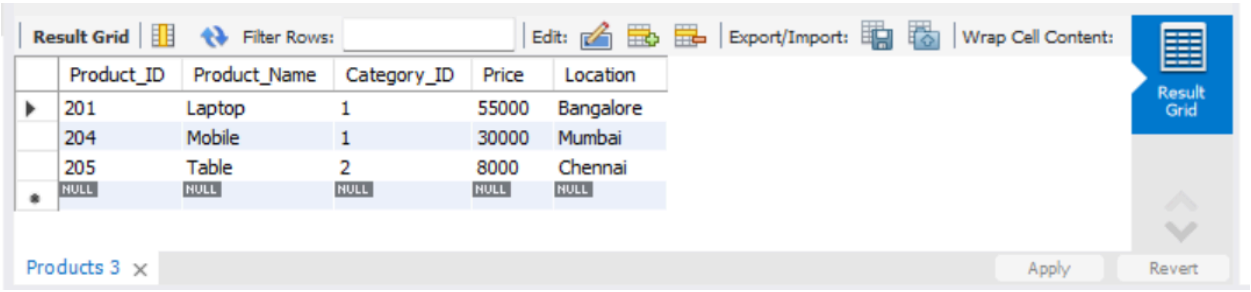
Products 1 | Apply | Revert

Subquery 5 — Products costing above the average Electronics price

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Price >
(SELECT AVG(Price) FROM Products WHERE Category_ID = 2);
```

OUTPUT:

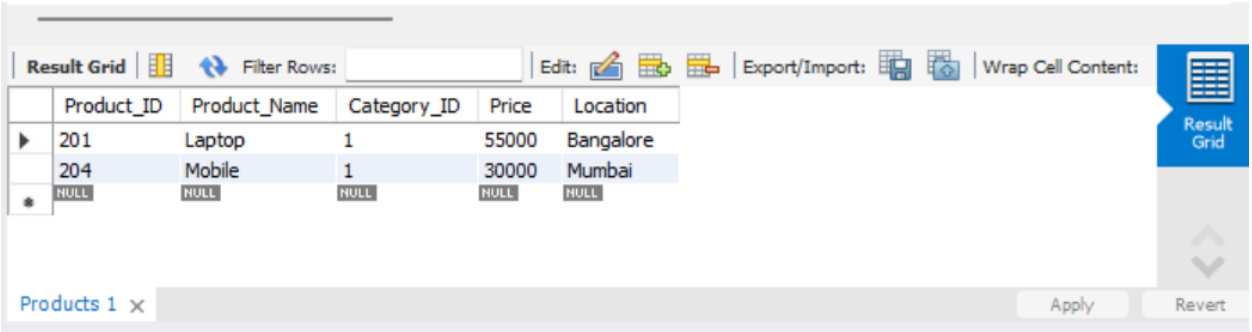


Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
205	Table	2	8000	Chennai
NULL	NULL	NULL	NULL	NULL

Subquery 6 — Products belonging to a category having a product from Mumbai

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Category_ID IN
(SELECT Category_ID FROM Products WHERE Location = 'Mumbai');
```

OUTPUT:



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

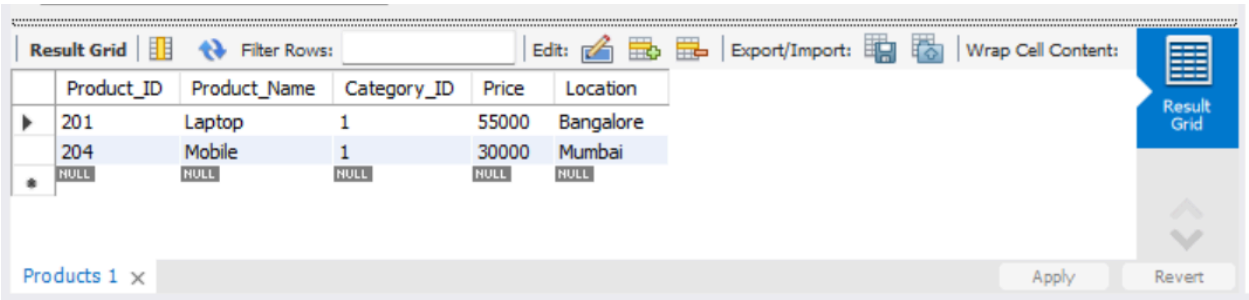
Products 1 | Apply | Revert

Subquery 7 — Products costing higher than every Furniture product

```sql
CREATE DATABASE IF NOT EXISTS StoreDB;
USE StoreDB;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
CREATE TABLE Categories (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50)
);
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category_ID INT,
    Price INT,
    Location VARCHAR(50)
);
```

```sql
INSERT INTO Categories (Category_ID, Category_Name) VALUES
(1, 'Electronics'),
(2, 'Furniture'),
(3, 'Stationery');
INSERT INTO Products (Product_ID, Product_Name, Category_ID, Price, Location) VALUES
(201, 'Laptop', 1, 55000, 'Bangalore'),
(202, 'Chair', 2, 4500, 'Mysore'),
(203, 'Notebook', 3, 120, 'Delhi'),
(204, 'Mobile', 1, 30000, 'Mumbai'),
(205, 'Table', 2, 8000, 'Chennai');
SELECT Product_ID, Product_Name, Category_ID, Price, Location
FROM Products
WHERE Price > (
    SELECT MAX(Price)
    FROM Products
    WHERE Category_ID = 2
);
```

OUTPUT:



Product_ID	Product_Name	Category_ID	Price	Location
201	Laptop	1	55000	Bangalore
204	Mobile	1	30000	Mumbai
NULL	NULL	NULL	NULL	NULL

Github:
https://github.com/manyappoojarybca25-byte/Dbms.git